

TopoMedKD: Shape- and Topology-Aware Distillation of Zero-Shot MedSAM for Compact Medical Image Segmentation

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Abstract. MedSAM offers a strong prior for medical image segmentation, but its transformer encoder and prompt-based inference make it difficult to deploy as an automatic model in constrained clinical settings. We propose TopoMedKD, a distillation framework that uses frozen, zero-shot MedSAM that is prompted with GT bounding box only during training and transfers its knowledge to a compact, automatic prompt-free student at inference time. The method combines three complementary modules: logit distillation, prototype-guided contrastive distillation, and a shape- and topology-aware structural loss based on signed-distance, soft-skeleton, and boundary consistency. A per-sample Focal-Tversky and Dice loss remains the dominant supervision term, limiting the risk of copying errors from the zero-shot teacher. On KiTS19 kidney tumour CT segmentation, TopoMedKD consistently improves lightweight students over supervised training, with the largest gain observed for ShufflenetV2. The results suggest that a frozen foundation model can be useful for efficient medical segmentation when the transfer objective is designed to preserve both class structure and region shape.

Keywords: Knowledge distillation · MedSAM · Efficient segmentation · Prototype-guided contrastive loss · Topology-aware loss

1 Introduction

Accurate lesion segmentation is important for treatment planning, follow-up, and quantitative image analysis. Recent foundation segmentation models, especially SAM [2] and MedSAM [1], have shown impressive generalisation across anatomical structures. Their practical use is still limited in two ways: they are computationally heavy, and they usually require prompts such as bounding boxes at inference time. This is a poor fit for automatic deployment on modest hardware or in high-throughput clinical pipelines.

Knowledge distillation can reduce this gap by training a small student to imitate a stronger teacher [3]. Existing MedSAM compression methods mainly reduce the encoder size or match features with mean-squared error [5,17,18,19], sometimes after a parameter-efficient teacher adaptation step [14]; medical KD more broadly has explored prediction-map and region-affinity transfer [4], and

prototype-based distillation [20] or cross-image relations [21] for general semantic segmentation. These strategies are useful, but they do not fully explain how the teacher separates foreground from background in feature space, nor do they directly transfer the shape of the predicted lesion. This matters for small tumours, where foreground pixels may be below one percent in many slices and a student can easily converge to an empty or fragmented mask.

We address this problem with **TopoMedKD**. During training, a frozen MedSAM teacher is prompted with the ground-truth bounding box to provide a supervision signal. At test time, the teacher and the prompt are removed; the student is fully automatic. The central idea is not to rely on the teacher prediction alone, since a zero-shot teacher can be imperfect, but to extract structured information that complements the ground-truth mask.

Our contributions are threefold. First, we formulate a prompt-free distillation pipeline from zero-shot MedSAM to compact CNN students. Second, we introduce Prototype-Guided Contrastive Distillation (PGCD), which uses teacher class prototypes as cross-model contrastive anchors for student pixel features, building on prototypical [9] and contrastive [10] learning. Third, we use Shape- and Topology-Aware Distillation (TGD), which transfers signed-distance, skeleton, and boundary cues from the teacher while keeping the supervised loss dominant. Experiments on KiTS19 show consistent gains across ESPNet, ENet, ShuffleNetV2, and ERFNet.

2 Method

2.1 Overview

Given a CT slice $\mathbf{x} \in \mathbb{R}^{1 \times H \times W}$ and a binary one-hot mask $\mathbf{y} \in \{0, 1\}^{2 \times H \times W}$, we train a compact student S with supervision from a frozen MedSAM teacher T . The teacher produces logits and two adapted feature maps,

$$(\mathbf{z}_T, \mathbf{F}_T^l, \mathbf{F}_T^h) = T(\mathbf{x}), \quad (\mathbf{z}_S, \mathbf{F}_S^l, \mathbf{F}_S^h) = S(\mathbf{x}), \quad (1)$$

where $\mathbf{z} \in \mathbb{R}^{2 \times H \times W}$ and $\mathbf{F}^l, \mathbf{F}^h$ denote feature maps at lower spatial resolutions. MedSAM weights are never fine-tuned. The teacher receives a ground-truth bounding box only during training, in order to elicit logits and features. The box is the tight bounding rectangle of the GT tumour mask; only tumour-bearing slices enter training and evaluation. The student never receives a box and is prompt-free at inference. Two lightweight teacher-side adapters project the MedSAM encoder output to the feature dimensions used for distillation. These adapters are trained jointly with the student; the MedSAM backbone remains frozen. The two adapters add only ~ 0.23 M trainable parameters. The total loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \alpha \mathcal{L}_{\text{LD}} + \beta \mathcal{L}_{\text{PGCD}} + \gamma \mathcal{L}_{\text{TGD}}, \quad (2)$$

with $\alpha = 0.1$ and $\beta = \gamma = 0.05$. The small auxiliary weights are intentional: the ground-truth loss remains the main signal, while the teacher regularises the

student toward useful feature and shape structure. The GT-derived bounding box and the prototype masks are privileged information available only at training and never at inference. Because the supervised baseline is trained on the same ground-truth masks, the supervised-vs-TopoMedKD comparison (Table 1) and the ablation (Table 3) isolate the teacher’s contribution *beyond* this label-derived supervision rather than adding extra labels. Figure 1 summarises the pipeline.

2.2 Logit Distillation (LD)

LD aligns output distributions via temperature-scaled KL [3]:

$$\mathcal{L}_{LD} = \tau^2 \text{KL}(\sigma(\mathbf{z}_S/\tau) \parallel \sigma(\mathbf{z}_T/\tau)), \quad \tau = 4, \quad (3)$$

with σ the softmax. Softening exposes the teacher’s relative class confidence; LD constrains *what* the teacher outputs per pixel.

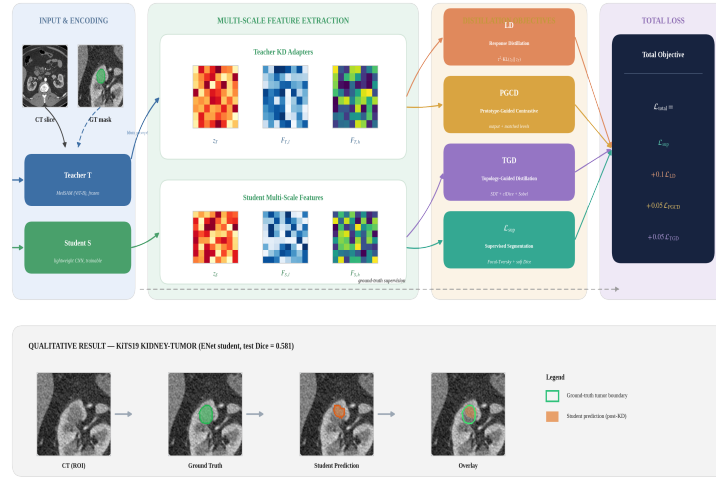


Fig. 1. Overview of TopoMedKD. Frozen MedSAM is prompted only during training. The compact student is trained with supervised segmentation loss plus logit, prototype-guided, and shape/topology-aware distillation.

2.3 Prototype-Guided Contrastive Distillation (PGCD)

PGCD recovers the teacher’s feature-space structure using per-class prototypes as cross-model anchors. Let Ω index the pixels of a feature level. For class $c \in \{0, 1\}$ and detached teacher feature $\mathbf{F}_T$, the prototype is the masked average

$$\mathbf{p}_T^c = \frac{\sum_{i \in \Omega} \mathbf{F}_T(i) \mathbf{1}[y_i = c]}{\sum_{i \in \Omega} \mathbf{1}[y_i = c] + \epsilon}. \quad (4)$$

After ℓ_2 -normalisation of the prototypes and of each student pixel feature $\hat{\mathbf{f}}_S(i) = \mathbf{F}_S(i)/\|\mathbf{F}_S(i)\|_2$, PGCD minimises a pixel-level contrastive cross-entropy:

$$\mathcal{L}_{\text{PGCD}} = -\frac{1}{|\Omega|} \sum_{i \in \Omega} \log \frac{\exp(\hat{\mathbf{f}}_S(i) \cdot \mathbf{p}_T^{y_i}/\eta)}{\sum_{k \in \{0,1\}} \exp(\hat{\mathbf{f}}_S(i) \cdot \mathbf{p}_T^k/\eta)}, \quad (5)$$

with $\eta = 0.07$ at the output level and $\eta = 0.1$ at feature levels. PGCD is applied at the output and any feature level where student and teacher channels match; mismatched levels are skipped.

2.4 Topology Guided Distillation (TGD)

LD and PGCD align outputs and features pixel-wise but leave the predicted region’s shape unconstrained. TGD matches three differentiable transforms of the foreground probability maps $p_S, p_T \in [0, 1]$, spanning geometry, connectivity, and boundary. We group these as shape- and topology-aware: the signed-distance and Sobel terms capture geometry and boundary location, while the soft-skeleton (clDice) term captures connectivity, the genuinely topological component.

Signed-distance surrogate (geometry): A hard Euclidean signed-distance transform requires thresholding the probability into a mask $M = \mathbf{1}[p > \frac{1}{2}]$, which is non-differentiable. To keep the term trainable end-to-end we instead use a smooth signed-distance *surrogate* of the foreground probability $p \in [0, 1]$,

$$\Phi(p) = (1 - 2p) \nu, \quad \nu = 6, \quad (6)$$

whose sign encodes inside ($\Phi < 0$ for $p > \frac{1}{2}$) versus outside and whose magnitude grows away from the $p = \frac{1}{2}$ level set, truncated to a ν -px band. Because Φ is affine in p , gradients reach the student directly through p_S with no thresholding on the gradient path; the teacher term is detached (stop-gradient $\text{sg}[\cdot]$):

$$\mathcal{L}_{\text{SDT}} = \text{SmoothL1}(\Phi(p_S), \text{sg}[\Phi(p_T)]). \quad (7)$$

This transfers the teacher’s interior extent and boundary placement beyond mere overlap while remaining fully differentiable in the student. This is the λ_d term of Eq. (10).

Soft clDice (connectivity): Following [6], let $\text{skel}(\cdot)$ be a differentiable soft-skeletonisation operator (iterative min/max-pooling). With teacher and student foreground maps p_T, p_S ,

$$\mathcal{L}_{\text{clD}} = 1 - \frac{2 T_{\text{prec}} T_{\text{sens}}}{T_{\text{prec}} + T_{\text{sens}}}, \quad T_{\text{prec}} = \frac{\sum \text{skel}(p_S) \odot p_T}{\sum \text{skel}(p_S)}, \quad T_{\text{sens}} = \frac{\sum \text{skel}(p_T) \odot p_S}{\sum \text{skel}(p_T)} \quad (8)$$

where $\odot$ is the element-wise product and sums run over pixels; T_{prec} measures how well the student skeleton lies inside the teacher region and T_{sens} the converse. We apply clDice as a distillation term (teacher map as reference) rather than against the binary mask. Although kidney tumours are compact rather

than tubular, soft-skeleton agreement still acts as a connectivity-aware regularizer against the two dominant failure modes under an imperfect teacher—fragmentation into disconnected blobs and erosion into pieces—by matching the teacher’s single-component skeleton, discouraging satellite components and interior holes that overlap losses do not penalise. **Sobel boundary alignment (boundary)**: Aligns gradient magnitudes so the student places its boundary where the teacher does:

$$\mathcal{L}_{\text{bnd}} = \left\| |\nabla p_S| - |\nabla p_T| \right\|_1. \quad (9)$$

The three combine linearly:

$$\mathcal{L}_{\text{TGD}} = \lambda_d \text{SmoothL}_1(\Phi(p_S), \Phi(p_T)) + \lambda_c \mathcal{L}_{\text{cID}} + \lambda_b \mathcal{L}_{\text{bnd}}, \quad (10)$$

with $\lambda_d = \lambda_c = 0.9$, $\lambda_b = 0.5$. TGD is applied at the output and the two feature levels; at feature levels, p_S, p_T are pseudo-probabilities from channel-mean saliency standardisation, and the teacher map is bilinearly resized to the student resolution. Concretely, for a feature map $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$ we form a saliency map $s = \frac{1}{C} \sum_c \mathbf{F}_c$, standardise it over spatial locations as $\hat{s} = (s - \mu_s) / (\sigma_s + \epsilon)$, and set the foreground pseudo-probability $p = \sigma(2\hat{s})$ (equivalently a softmax over the pseudo-logits $[-\hat{s}, +\hat{s}]$).

2.5 Supervised Loss

Tumour foreground occupies less than one percent of the image in many slices; batch-averaged losses can collapse to the empty mask. We use a per sample Focal Tversky [8,7] combined with soft Dice:

$$\mathcal{L}_{\text{sup}} = w_{ft} (1 - \text{Tversky})^\rho + w_d \mathcal{L}_{\text{Dice}}, \quad (11)$$

with Tversky weights $\alpha_T = 0.3 < \beta_T = 0.7$ (penalising missed tumours more than false positives), focusing exponent $\rho = 4/3$, and $w_{ft} = 0.6$, $w_d = 0.4$. Computing both terms per sample and averaging keeps the foreground gradient informative regardless of lesion size. The auxiliary weights in Eq. (2) keep $\mathcal{L}_{\text{sup}}$ dominant, so ground truth, not the teacher, anchors training.

3 Experiments

3.1 Setup

We evaluate on KiTS19 [11] CT kidney-tumour segmentation. The cases are partitioned 70%/15%/15% into training, validation, and test sets (case-wise, so that all slices of a patient remain within a single split). We distil into lightweight students spanning two orders of parameter magnitude ESPNet [15], ENet [12], ShuffleNetV2 [16], and ERFNet [13] and report Dice, VOE, and surface HD95 in pixels, averaged per case; KD weights follow Eq. (2). For every student we compare a supervised-only baseline (no teacher) against the full TopoMedKD

pipeline (Table 1). All models are implemented in PyTorch and trained on a single GeForce RTX 3080 Ti (16 GB). The student is optimised with Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$) and batch size 16, at an initial learning rate of 1×10^{-3} annealed by a cosine schedule ($\eta_{\min} = 1 \times 10^{-6}$) over 100 epochs; the MedSAM teacher (public ViT-B checkpoint `medsam_vit_b`) stays frozen throughout and only its two lightweight adapters are updated jointly with the student.

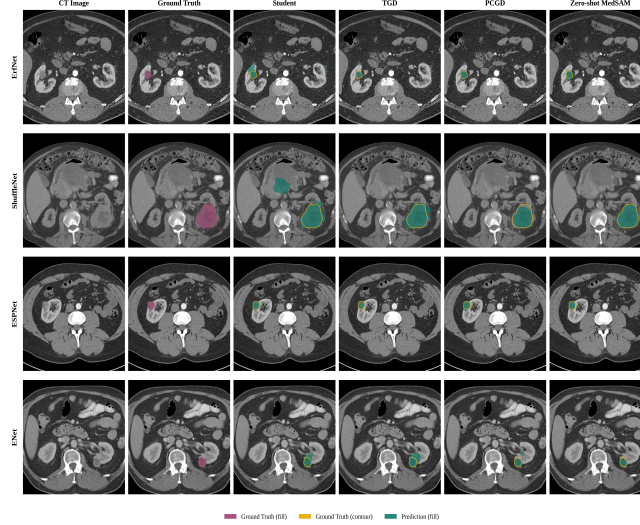


Fig. 2. Qualitative comparison of segmentation predictions. Ground truth is shown in purple (fill) and yellow (contour); predictions are overlaid in green.

3.2 Results

Table 1 reports TopoMedKD against the supervised baseline for each student under the zero-shot teacher. TopoMedKD improves every student over its supervised baseline by a substantial margin while leaving the teacher untouched. The largest absolute Dice gain is obtained by ShuffleNetV2, which moves from 0.466 to 0.647 and cuts HD95 from 155.7 to 80.1 px; ERFNet and ESPNet show comparable improvements. The smallest relative gain is observed on ENet, the student that already benefits from the strongest supervised baseline (0.506). The pattern is informative: the weaker the student’s supervised prior, the more it gains from the structured KD signal, which is the expected behaviour of distillation acting as a regulariser of an under parameterised model. Boundary localisation improves in lockstep with region overlap. HD95 is more than halved for ShuffleNetV2 (−75.6 px) and ERFNet (−58.7 px), and approximately halved for ESPNet (−43.8 px); this is a regime where supervised CT-only training typically saturates because the Focal Tversky loss has no explicit shape term. VOE

drops in the same direction across all four students, confirming that the gain is not a side-effect of biased overlap on small foreground regions. Notably, all four results are obtained from a frozen MedSAM teacher that has never seen KiTS19 data, indicating that the zero-shot prior is rich enough to support meaningful transfer once the right structural signal is extracted from it.

Table 1. KiTS19 tumour segmentation under a zero-shot MedSAM teacher.

Student	Params	GFlops	Method	Dice $\uparrow$	VOE $\downarrow$	HD95 $\downarrow$
ESPNet	0.18	1.27	Supervised	0.412	0.592	124.1
			TopoMedKD	0.538	0.307	80.3
ENet	0.35	2.20	Supervised	0.506	0.401	66.4
			TopoMedKD	0.581	0.259	60.4
ShuffleNetV2	1.18	4.18	Supervised	0.466	0.659	155.7
			TopoMedKD	0.647	0.375	80.1
ERFNet	2.06	16.57	Supervised	0.489	0.630	120.1
			TopoMedKD	0.655	0.281	61.4

Figure 2 overlays representative KiTS19 test predictions (green) on the CT slice together with the ground truth (purple fill, yellow contour). The qualitative behaviour matches the quantitative trends in Table 1: the distilled masks enclose the tumour tightly, with the green prediction tracking the yellow contour rather than bleeding into adjacent kidney parenchyma, and they remain connected as a single region instead of fragmenting the failure mode explicitly targeted by the soft-skeleton (cIDice) and signed-distance terms of TGD. Agreement is tightest on well-defined lesions and hardest on small or low-contrast ones, the regime where imbalance-aware supervision alone tends to under-segment; the residual discrepancies concentrate almost entirely on the lesion boundary, consistent with HD95 remaining the hardest metric to saturate.

3.3 Ablation

The ablation in Table 3 isolates the contribution of each term and shows that the three signals are genuinely complementary rather than redundant. Adding LD to the supervised baseline accounts for the largest single jump (+0.100 Dice, -58.5 px HD95) and confirms that the zero-shot teacher’s softened posterior already carries useful information beyond the GT mask. PGCD and TGD add further but distinct gains. PGCD on top of LD contributes +0.049 Dice and -6.3 px HD95, acting in feature space by anchoring student pixels to the teacher’s class prototypes. TGD on top of LD contributes +0.040 Dice and -1.9 px HD95, acting in output space by aligning shape, connectivity, and boundary descriptors. The two terms address different deficiencies of LD-PGCD constrains *what* representation the student forms, TGD constrains *how* the prediction is shaped and

combining them recovers $+0.032$ Dice and -10.8 px HD95 beyond LD + PGCD, reaching the final 0.647 Dice and 80.1 px HD95. The fact that the gains stack rather than saturate validates the design choice of using three signals at three levels of structure (logits, features, shape descriptors). Figure 3 plots the four loss components over training for each student, confirming that all terms decrease smoothly and remain stably balanced under the fixed weighting of Eq. (2). Replacing the zero-shot teacher with a fine-tuned MedSAM under the same TopoMedKD objective produces a more nuanced picture than usually reported for teacher adaptation. On ENet, the fine-tuned teacher lowers HD95 from 60.4 to 43.7 px but also lowers Dice from 0.581 to 0.546. This dissociation is consistent with the way adaptation reshapes the teacher signal: fine-tuning produces sharper, more KiTS19 aligned boundaries which transfer cleanly through TGD’s shape and Sobel terms and explain the HD95 improvement but it

Table 2 compares zero-shot and fine-tuned teachers under the full TopoMedKD objective on ENet, isolating teacher adaptation from the distillation modules.

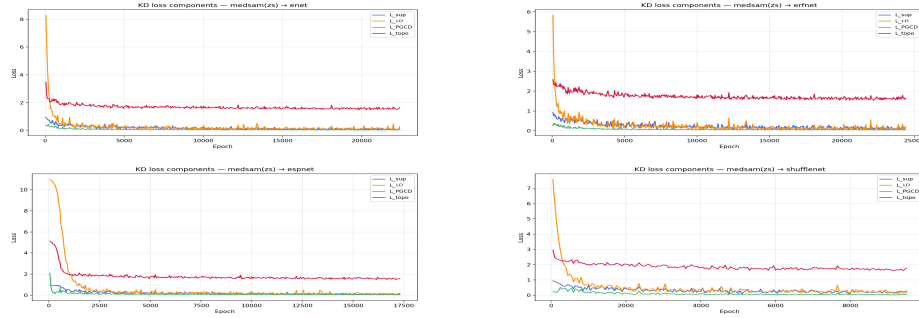


Fig. 3. KD loss components during training for each student under the zero-shot MedSAM teacher.

Table 2. Effect of teacher adaptation with full TopoMedKD on ENet. The value in parentheses after each teacher is that teacher’s *own* zero-shot / fine-tuned Dice on KiTS19 (teacher quality); the Dice and HD95 columns report the resulting *student*.

Teacher	Dice $\uparrow$	HD95 $\downarrow$	Teacher train cost
Zero-shot MedSAM (0.799)	0.581	60.4	none
Fine-tuned MedSAM (0.762)	0.546	43.7	encoder adaptation

also narrows the posterior and the per-class prototypes, removing some of the soft inter-class structure that PGCD and LD rely on for their gradients. The result is a teacher that is locally more precise but globally less informative for distillation, an effect already observed in other KD settings where over-confident

teachers degrade transfer. In our cost regime the trade-off favours the zero-shot configuration: +0.035 Dice at the price of +16.7 px HD95, with zero GPU-hours spent on teacher adaptation, no labels consumed on the teacher side, and a guarantee that the reported gains are attributable to the distillation modules rather than to teacher specialisation. This supports the design choice of treating MedSAM as a fixed prior and concentrating modelling effort on the transfer signal.

Table 3. Module ablation on ShuffleNetV2 with the zero-shot teacher. The top row is supervised-only training(no KD).

LD	PGCD	TGD	Dice $\uparrow$	HD95 $\downarrow$
			0.466	155.7
✓			0.566	97.2
✓		✓	0.606	95.3
✓	✓		0.615	90.9
✓	✓	✓	0.647	80.1

4 Conclusion

TopoMedKD shows that a frozen MedSAM teacher can improve compact, prompt-free students when its knowledge is transferred through more than pixel-wise response matching. The combination of logit, prototype-guided, and shape/topology-aware distillation improves four lightweight networks on KiTS19. Consistent with prior KiTS distillation setups, training and evaluation here use tumour-bearing slices only; this isolates segmentation quality but does not measure false positives on negative slices or over whole volumes, so full-volume, false-positive-aware evaluation is an important next step toward the fully automatic clinical use we target. Extending to further organs and modalities would better establish generality. Finally, the connectivity effect of the cIDice term is argued qualitatively; quantifying it with connected-component counts and hole statistics, together with volumetric (3D) topology metrics and box-free teacher prompting, would strengthen the topological claim.

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